

Titanic Dataset Preprocessing Practical

Aim

To perform preprocessing and visualization on the Titanic dataset using Python libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn.

Objectives

- Load and analyze the Titanic dataset.
 - Handle missing values.
 - Remove unnecessary columns.
 - Convert categorical data into numerical format.
 - Perform feature scaling.
 - Visualize the dataset using graphs and plots.
 - Save the processed dataset for Machine Learning tasks.
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Theory

Data preprocessing is an important step in Machine Learning and Data Science. Raw datasets usually contain missing values, duplicate records, categorical data, and outliers. These issues affect the performance of Machine Learning models.

The Titanic dataset contains passenger information such as age, gender, fare, ticket class, and survival status. In this practical, preprocessing techniques are applied to clean and transform the dataset into a machine-learning-ready format.

Libraries Used

- Pandas – Data manipulation and analysis
 - Seaborn – Statistical visualization
 - Matplotlib – Plotting graphs and charts
 - Scikit-learn – Feature scaling and preprocessing
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Preprocessing Techniques Used

- Missing Value Handling
 - Feature Scaling
 - One Hot Encoding
 - Duplicate Removal
 - Data Visualization
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Formulas Used

Mean Formula:

Mean = Sum of Values / Total Number of Values

Standardization Formula:

$Z = (X - \text{Mean}) / \text{Standard Deviation}$

Visualizations Included

- Survival Count Plot
 - Age Distribution Histogram
 - Fare Boxplot
 - Correlation Heatmap
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Code Explanation

1. The dataset is loaded using Pandas.
 2. Dataset information and missing values are displayed.
 3. Graphs and plots are generated for visualization.
 4. Missing values are replaced using mean and mode.
 5. Unnecessary columns are removed.
 6. Categorical values are converted into numerical form.
 7. Duplicate records are removed.
 8. Numerical features are standardized using StandardScaler.
 9. The processed dataset is saved as a CSV file.
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How to Run the Practical in Anaconda (.py File)

1. Open Anaconda Navigator.
 2. Launch Spyder or Jupyter Notebook.
 3. Install required libraries using:
`pip install pandas seaborn matplotlib scikit-learn`
 4. Save the Python code as Titanic_Preprocessing.py
 5. Place the dataset file in the same folder.
 6. Run the program using the Run button or terminal command:
`python Titanic_Preprocessing.py`
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How to Run the Practical in Anaconda (.ipynb File)

1. Open Anaconda Navigator.
2. Launch Jupyter Notebook.
3. Open the .ipynb notebook file.

4. Ensure the dataset CSV file is in the same folder.
 5. Run cells one-by-one using Shift + Enter.
 6. All graphs and outputs will be displayed automatically.
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Expected Output

- Cleaned Titanic dataset
 - Graphical visualizations
 - Processed CSV file
 - Machine-learning-ready dataset
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Outcome

Successfully performed preprocessing and visualization on the Titanic dataset using Python. The dataset was cleaned, transformed, and prepared for Machine Learning applications.

Conclusion

This practical demonstrates the importance of preprocessing in Data Science and Machine Learning. By handling missing values, encoding categorical data, scaling features, and visualizing data, the dataset becomes suitable for analysis and predictive modeling.

Important Viva Questions with Answers

1. What is Data Preprocessing?

Data preprocessing is the process of cleaning and transforming raw data into a useful format.

2. Why is preprocessing important?

It improves data quality and increases Machine Learning model accuracy.

3. What is feature scaling?

Feature scaling standardizes numerical data into a common range.

4. What is one hot encoding?

It converts categorical data into binary numerical columns.

5. Why was the Cabin column removed?

Because it contained many missing values.

6. Which library is used for visualization?

Matplotlib and Seaborn.

7. What is the purpose of StandardScaler?

It standardizes data using mean and standard deviation.

8. What is a heatmap?

A heatmap visually represents correlations or missing values in a dataset.

9. What is the use of Pandas?

Pandas is used for data manipulation and analysis.

10. What is the output of this practical?

A cleaned and processed Titanic dataset ready for Machine Learning.
